MARÍA JOSÉ GÓMEZ QUIJANO
EEB Postdoctoral Fellow, University of Toronto
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GOOGLE SCHOLAR:

https://scholar.google.com/citations?user=7Vxn87kAAAAJ&hl=en

EDUCATION

Queen's University at Kingston, April 2026
Biology PhD, Colautti Lab
Outstanding Thesis Award

University of Minnesota Duluth. July 2021
Integrated Biosciences M.Sc., Gross Lab and Etterson Lab
GPA: 3.905

University of Minnesota, Duluth. May 2018
Cell and Molecular Biology B.S
Communication B.A (Cum Laude)
Graduating GPA: 3.66

RESEARCH EXPERIENCE

University of Toronto, Department of Ecology and Evolutionary Biology. **May 2026 – Present**
Postdoctoral Fellow
PI: Dr. Stephen Wright
Project description: My research incorporates population and temporal genomics to reconstruct the introduction history and understand rapid adaptation of *Arctotheca calendula* in Australia. This project uses historical herbarium specimens to track demographic and genomic changes through time between the invasive and the introduced range. I am also building phased chromosome-level genome assemblies to construct a capeweed pangenome, providing a genomic framework for investigating structural variation across populations.

Queen's University, Biology Department. 2021 - 2026
PhD Biology Program
PI: Dr. Robert Colautti
Thesis: **Investigating ecological and evolutionary processes that shape the evolutionary trajectories and demographic histories of widespread native and invasive populations along environmental gradients.**
Project Description: Using an integrative approach that combines quantitative (common garden experiment), molecular genetic techniques (RAD-seq and GT-seq) and focusing on native species in the genus *Calamagrostis* and the North American invasive plant *Lythrum salicaria*; My research investigated which evolutionary processes best explain population structure of *Calamagrostis* species used for ecological restoration in the Northwest Territories and the invasion history and genetic structure of *Lythrum salicaria* populations in Eastern North America. I also investigated the role of flowering time-

based assortative mating as a mechanism that facilitates rapid adaptation, and the effects of multigenerational flowering time and geography based non-random mating under environmental stress in *L. salicaria.*

University of Minnesota Duluth, Department of Biology. 2018 - 2021
MSc. Integrated Biosciences Program
PI: Dr. Briana L. Gross, Dr. Julie R. Etterson
Thesis: **Why Do Coastal Seeds Fail? Evidence of Local Adaptation of Northern Red Oak (*Quercus rubra*) in Minnesota Coastal Forests.**
Project Description: Using population, quantitative and landscape genetics, common garden experiments and focusing on Northern red oak trees (*Q. rubra*); My research investigated population structure, hybridization and molecular signatures of selection in coastal and inland Minnesota forests. My research also used past and present environmental data to identify correlations between climate factors and population differentiation due to the moderating effect of Lake Superior.

PUBLICATIONS

In press:

1. **Quijano MJG**, Wu Y, Smith C, López-Villalobos A, Secombe-Hett P, Campbell N, Sun Z, Colautti RI (2026). Genome-Wide SNP Diversity in Natural and Cultivated Populations Informs Restoration Ecology with Three Calamagrostis Species from the Northwest Territories of Canada. *Molecular Ecology*, 35(14), e70484. https://doi.org/0.1111/mec.70484

2. **Gómez Quijano, M. J.**, Gross, B. L., & Etterson, J. R. (2024). Genetic differentiation across a steep and narrow environmental gradient: Quantitative genetic and genomic insights into Lake Superior populations of *Quercus rubra. Molecular Ecology*, 33, e17483. https://doi.org/10.1111/mec.17483

Preprints:

1. **Gomez Quijano, M. J**., Akbar, M. N., Crowson, E., Colautti, R.I. (2026). "Assortative mating may have facilitated rapid evolution of locally adaptive phenologies during a biological invasion." bioRxiv. 2026.10.08.757682; https://doi.org/10.64898/2026.10.08.757682

TEACHING EXPERIENCE

Teaching Fellow / Co-instructor on record with PhD candidate Alyson Van Natto
Queen's University of Kingston, Biology Department – Fall 2025
Course: **BIOL 860 – Introduction to Data Management and Statistical Analysis of Biological Data**
Work Description: Co-develop and deliver lectures two times a week to first year graduate students. Topics include experimental design, hypotheses testing with appropriate statistical analyses, parameter estimation, and graphing.in R studio. Design and grade assignments, quizzes, and exams to evaluate student comprehension and performance throughout the semester. Manage all course logistics and administration, including OnQ (D2L), QSAS accommodations, and Queen's Exams Office.

Teaching Fellow / Co-instructor on record with PhD candidate Drew Sauve
Queen's University of Kingston, Biology Department - Winter 2023
Course: **BIOL 206 - Evolutionary Genetics**.
Work Description: Co-developed and delivered lectures three times a week to ~270 second-year undergraduate students. Taught core concepts in micro- and macroevolution, ecological genetics, population genetics, and quantitative genetics. Designed and graded assignments, quizzes, and exams to evaluate student comprehension and performance throughout the semester. Managed all course logistics and administration, including OnQ (D2L), QSAS accommodations, and Queen's Exams Office.

Guest Lecture for BIOL 343 - Data Analysis for Biologists
Queen's University of Kingston, Biology Department - Fall 2024
Instructor: Dr. Robert Colautti
Work Description: Developed and instructed a 1.5hr guest lecture on advanced linear models including topics as interactions, collinearity and model performance. Developed PowerPoint materials and in class worked exercise in R studio.

Guest Lecture for BIOL 343 - Data Analysis for Biologists
Queen's University of Kingston, Biology Department - Fall 2022
Instructor: Dr. Robert Colautti
Work Description: Instructed a 2.5hr guest lecture on statistical model selection that included topics such as minimum adequate model selection techniques, likelihood ratio tests and Information criteria estimates. Developed PowerPoint materials and in class worked exercise in R studio.

Graduate Teaching Assistant for BIOL 343 - Data Analysis for Biologists
Queen's University of Kingston, Biology Department - Fall 2021, 2022, 2023, 2024
Instructors: Dr. Robert Colautti (Fall 2021, 2022, 2024), Dr. Chris Eckert (Fall 2023)
Work Description: Instructed tutorials and supported students with weekly assignments, code troubleshooting and course questions. Topics included basic R studio, use of the tidy verse, creating publication quality plots and graphs using ggplot, managing large datasets, data wrangling, fundamental statistical concepts (i.e. distribution, central moments), introductory and advanced statistical tests and models (i.e. t-test, linear models, linear mixed models, polynomial regression, generalized linear models) and experimental design for biologists. Graded course assignments, exams, and quizzes. Development two student assignments on basic R usage and experimental design and developed grading rubrics.

Graduate Teaching Assistant for BIOL 432 - Computation, Big Data in Biology
Queen's University of Kingston, Biology Department - Winter 2022, 2025
Instructor: Dr. Robert Colautti
Work Description: Instructed Tutorials, focusing on the usage of computational methods to analyze big datasets in biology. Topics included supervised and unsupervised machine learning algorithms, phylogenetic analyses, sequence alignment, genome assembly, transcriptomics and advanced R studio and introduction to HPCs. Graded course assignments, exams, and quizzes. Assisted in the development of grading rubrics.

Graduate Teaching Assistant for BIOL206 – Evolutionary Genetics
Queen's University of Kingston, Biology Department – Winter 2024, 2025
Course Instructor: Dr. Howard Teresinski
Work Description: Instructed tutorials, focusing on the application of evolutionary genetics concepts through case studies and in class activities. Graded course assignments and guided students through course material.

Graduate Teaching Assistant for BIOL307 – Field Biology
Queen's University of Kingston, Biology Department - Summer 2024
Instructor: Dr. Sarah Yakimowski
Work Description: Instructed an introductory tutorial to statistical analysis, focusing on the use of linear models for independent project development. Supported students through field trips, project planning, plant ID and mounting herbarium specimens.

Graduate Teaching Assistant for BIOL 102 - Fundamentals of Biology: Molecular and Cell Biology
Queen's University of Kingston, Biology Department - Fall 2023
Course Instructor: Dr. Barbara Vanderbeld
Work Description: Instructed labs, focusing on the application of introductory molecular and cell biology concepts through case studies and in class activities. Graded course assignments and guided students through course material.

Graduate Teaching Assistant for General Biology I
University of Minnesota Duluth, Department of Biology. Fall 2018. Spring 2019
Instructors: Dr. Joanne Itami (Fall 2018), Dr. Shannon Stevenson (Spring 2019).
Work Description: Instructed laboratories and discussion sections, focusing on the process of science, introductory biology, and science writing. Developed class activities, graded laboratory activities, discussions and writing exercises. Assisted course instructor in Lectures.

Graduate Teaching Assistant for Molecular Biology
University of Minnesota Duluth, Department of Biology - Spring 2020
Course Instructor: Dr. Jessica Sieber
Work Description: Instructed laboratories using in person and online modalities, focusing on molecular and microbiology techniques and science writing. Developed teaching materials, graded laboratory activities and written reports. Assisted Course instructor.

Graduate Teaching Assistant for Genetics
University of Minnesota Duluth, Department of Biology - Fall 2020
Course Instructor: Dr. Katie Nemeth
Work Description: Instructed laboratories using in person and online modalities, focusing on mendelian genetics concepts, molecular genetic techniques, and science writing. Developed teaching materials, graded laboratory activities and written reports. Assisted Course instructor.

WORKSHOP DEVELOPMENT

Workshop teaching and development - Absolute Zero to Coding Hero
Canadian society of Ecology and Evolution (CSEE) conference, Toronto, ON - 2026
Work Description: Worked alongside a team of graduate students and Dr. Rob Colautti to develop and instruct a beginner friendly R coding workshop. This course covered the fundamentals of R studio, data wrangling, basic graphical methods, and introduction to community ecology analyses. For this course I developed materials on the fundamentals of R studio and data wrangling and taught the workshop to ~ 20 people.

Workshop teaching and development - Absolute Zero to Coding Hero
Canadian society of Ecology and Evolution (CSEE) conference, Winnipeg, MB - 2023
Work Description: Worked alongside a team of graduate students and Dr. Rob Colautti to develop and

instruct a beginner friendly R coding workshop. This course covered the fundamentals of R studio, data wrangling, basic graphical methods, basic linear model and an introduction to machine learning algorithms. For this course I developed materials on graphic methods, machine learning and multivariate analysis and taught 2 sections of the workshop to ~ 40 people.

Workshop teaching and development - Absolute Zero to Coding Hero
Ecological Society of America (ESA) conference, Online - 2023
Work Description: Worked alongside a team of graduate students and Dr. Rob Colautti to develop and instruct a beginner friendly R coding workshop. This course covered the fundamentals of R studio, data wrangling, basic graphical methods and basic linear model. For this workshop I taught 4 sections of the workshop to ~30 people online and helped with troubleshooting and one-on-one online questions.

RESEARCH ASSISTANTSHIPS

University of Minnesota Duluth, Institute on the Environment. Spring 2021
The Institute on The Environment - Duluth
Supervisor: Dr. Julie Etterson
Work Description: Coordinated and facilitated weekly panel discussions on atopic related to the 17 United Nations Sustainable Development Goals. The panels were created among requests from faculty and designed specifically for the class. I interacted directly with faculty and connected with community leaders, experts and scholars to request their participation in these panels. I also moderated the panels and facilitated interdisciplinary dialogue to foster sustainability action in the community.

PRESENTATIONS

National:

Oral paper: CSEE 2025, Canadian Society for Ecology and Evolution. Toronto, ON. May 2026
Topic: **Potential for assortative mating may facilitate evolution of locally adaptive phenologies during a biological invasion.**

Poster: CSEE 2023, Canadian Society for Ecology and Evolution, Winnipeg MB, June 2023
Topic: **Effects of multi-generational non-random mating under environmental stress in Purple Loosestrife (*Lythrum salicaria*).**

Oral paper: ESA/CSEE 2022, Ecological Society of America and Canadian Society for Ecology and Evolution. Montreal QC. August 2022
Topic: **Genomic Tools to Inform Restoration Strategies of Three Species of *Calamagrostis* in Canada's Northwest Territories.**

Oral paper: OE3C 2022, Ontario Ecology Ethology and Evolution Colloquium. Kingston ON, June 2022
Topic: **Genomic Tools to Inform Restoration Strategies of Three Species of *Calamagrostis* in Canada's Northwest Territories.**

International:

Symposium Organizer: Evolution 2024, Society for the Study of Evolution. Montreal, QC. August 2024
Topic: **The Politics of Citation in Evolutionary Biology**

Oral paper: Evolution 2024, Society for the Study of Evolution. Montreal, QC. August 2024
Topic: Who's the perfect match: Fitness **Effects of Non-Random Mating Under Environmental Stress in Purple Loosestrife (*Lythrum salicaria*).**

Poster: Evolution 2022, Society for the Study of Evolution, Cleveland, OH. June 2022
Topic: **Effects of Assortative Mating on Inbreeding and Outbreeding Depression Under Environmental Stress in Purple Loosestrife (*Lythrum salicaria*).**

Oral paper: Botany 2021, Botanical Society of America, online. July 2021
Topic: **Evidence of Local Adaptation of Northern Red Oak (*Quercus rubra*) in Minnesota Coastal Forests.**

Oral paper: Botany 2020, Botanical Society of America, online. July 2020
Topic: **Why Do Coastal Seeds Fail? Evidence of Local Adaptation of Northern Red Oak (*Quercus rubra*) in Minnesota Coastal Forests.**

Poster: Botany 2019, Botanical Society of America, Tucson, AZ. July 2019
Topic: **Why Do Coastal Seeds Fail? Evidence of Local Adaptation of Northern Red Oak (*Quercus rubra*) in Minnesota Coastal Forests.**

Poster: Our Climate Futures Conference, University of Minnesota Duluth. April 2019
Topic: **Why Do Coastal Seeds Fail? Evidence of Local Adaptation of Northern Red Oak (*Quercus rubra*) in Minnesota Coastal Forests.**

Poster: Botany 2018, Botanical Society of America, Rochester, MN. July 2018
Topic: **Variation in Cyanide Production of White Clover (*Trifolium repens*) Across a Steep Environmental Gradient.**

WORKSHOP PARTICIPATION AND PROFESSIONAL DEVELOPMENT

CIEE Working group, Temporal variation in dispersal - Toronto, ON. May 2025
Description: Invited to be part of an international working group to analyze existing data and literature on the effects of temporal variation is dispersal (masting) to evaluate how it can affect range dynamics.

Project Baseline Workshop, Evolution Conference 2022 – Cleveland OH. June 2022
Evolution Conference 2022
Course Description: Attended a networking and project development workshop, focused on using the project baseline seedbank to develop resurrection experiments.

RadCamp 2019 (IpyRAD Workshop – Part II), New York City, NY. October 2019
Columbia University

Course Description: Learned RADseq assembly, phylogenetic and population genetic methods using high performance computing, basic Unix command line and python programming. Workshop led by the developers of IpyRAD, Deren Eaton and Isaac Overcast.

Introduction to Reproducible Data Science with R, Minneapolis, MN. August 2019
University of Minnesota Twin Cities
Course Description: Learned the basic features of R and best practices for reproducible data science using R Markdown. Learned how to produce graphs, data visualizations and statistical models such as linear and logistic regression in R.

Teaching in the Life sciences (Biol 5001), Duluth, MN. Spring 2019
University of Minnesota Duluth
Course Description: Learned about learning theory and educational practices designed to help develop skills in the classroom. Course topics included learning styles, classroom management, assessment, active and cooperative learning, and educational technology.

MEDIA

1. CBS What on Earth: Meet 2 PhD students who are trying to make environmental field work more accessible: https://www.cbc.ca/news/science/what-on-earth-environmental-field-work-1.6824554
2. Climate change and trees – University of Minnesota Duluth, YouTube series: https://www.youtube.com/watch?v=1wpwymgUiVE

UNDERGRADUATE MENTORSHIP

Queen's University of Kingston, Department of Biology. 2024-2025
Biol 537 Thesis
Mentor and project co-supervisor
Student Mentored: **Jessica Smuck**
Project: **Impacts of Paternal Latitudinal Origin on Herbivore Response Strategies in *Lythrum salicaria***

University of Minnesota Duluth, Department of Biology. 2019-2020
Biology Undergraduate Research in Science and Technology (BURST) fellowship program
Mentor and project co-supervisor
Student Mentored: **Michael Laidlaw**
Project: **Morphological Variation in Minnesota's Northern Red Oak (*Quercus rubra*) Populations**

University of Minnesota Duluth, Department of Biology. 2020-2021
Undergraduate Research Opportunity Program (UROP)
Mentor and project co-supervisor
Student Mentored: **Samantha Ykema**
Project: **Response of *Quercus rubra* to a Predicted Change in Minnesota Climate**

LEADERSHIP EXPERIENCE

Queen's University, Kingston, ON. September 2022 - 2024
Biology Graduate Student Council
Position: **I-EDIAA graduate student representative**
Position Description: Attend regular council meetings, reported on I-EDIAA departmental initiatives and issues, and advocate for graduate students' needs and wellbeing. Promote equitable and just practices in the graduate student body and collate internal and external resources on best practices.

Queen's University, Kingston, ON. September 2022 – Present
Position: **Co-leader and Facilitator, Anti Oppression in STEM Workshops,**
Position description: Developed workshops with an interdisciplinary group of STEM graduate students to implement anti-oppression practices for graduate teaching assistants and faculty. Contributed to manuscript writing and grant applications to extend the workshops to the broader university community.

Queen's University, Kingston, ON. September 2021 – August 2022
Biology Graduate Student Council
Position: **International graduate student representative**
Position Description: Attended regular council meetings, reported on international student campaigns and advocated for international student wellbeing. Coordinated meetings with international students, directed them to relevant institutional resources within the university and government bodies.

Queen's University, Kingston, ON. April 2022 – September 2022
Biology Graduate Student Council and Biology Department
Position: **Organizer, Biology Graduate Research Symposium**
Position Description: Recruited graduate and keynote speakers to present at the annual biology research symposium. Prepared and oversaw conference budgets that included room bookings, food, logistics and merchandise for ~100 attendees. Designed and procured conference logo and merchandise.

Queen's University, Kingston, ON. May 2022 – 2024
Position: **Co-chair, Queen's Outdoor Field Experience Initiative,**
Position Description: Enable access to free outdoor gear and education for Queen's students through the organization and development of a gear lending library. Identify barriers to outdoor and field work experiences and develop strategies to counter systemic inequalities. Organize events to promote outdoor education and encourage usage of the gear library.

University of Minnesota, Twin Cities and Duluth, MN. October 2020 – May 2021
Institute on the Environment, Acara Graduate Changemakers Lab
Position: **Graduate leader and graduate program coordinator**
Position Description: Worked with a diverse group of graduate students from the University of Minnesota to develop and execute training programs focused on sustainability, environmental and social justice. Attend regular leadership and professional development training from the Institute on the environment.

City of Duluth Energy Plan Commission**,** Duluth, MN. November 2019 – August 2021 Position: **Commissioner and vice-chair**
Position Description: Worked with a diverse group of community leaders and experts to develop and

recommend policies consistent with the city's greenhouse gas reduction goals. Reviewed and evaluated the city's energy plan and assessed whether benchmarks are met. Promote the city's use of clean, renewable energy resources. Promote community best practices and strategies for reducing greenhouse gasses.

LANGUAGES

Native proficiency in Spanish
Bilingual/Native proficiency in English
Beginner proficiency in French

Programing languages:
Advanced proficiency in R studio
Advanced proficiency in Unix and HPC clusters
Beginner proficiency in Python

COMMUNITY VOLUNTEERING

Science Rendezvous, Presenter, Kingston, ON. May 2022,2023,2024
Northeastern Minnesota Science Fair Judge, Duluth, MN. February 2017,2018,2019, 2020
Lake Superior youth Symposium, Duluth, MN. May 2019
Office of Diversity and Inclusion, tutor and mentor, Duluth MN. 2015-2017

AWARDS

Type: Departmental Award/scholarship
Title: E. W. Derbyshire Award
Amount: 6,000 CAD
Institution: Queen 's University. December 2024

Type: Departmental Award
Title: Biology Department – I-EDIAA champion award
Amount: 500 CAD
Institution: Queen's University. March 2023

Type: Departmental Award
Title: Biology Department – Volunteer award
Amount: 100 CAD
Institution: Queen's University. March 2023

Type: Conference Support
Title: Biology Department – Travel Award
Amount: 250 CAD
Institution: Queen's University. August 2022

Type: Conference Support
Title: Evolution Conference 2022 – Student Travel Award
Amount: 500 USD
Institution: Society for the Study of Evolution. June 2022

Type: Conference Support
Title: Evolution Conference 2022 – Project Baseline Travel Award
Amount: 350 USD
Institution: Project Baseline. June 2022

Type: Conference Support
Title: Department of Biology – Travel Award
Amount: 500 USD
Institution: University of Minnesota Duluth. August 2019

Type: Conference Support
Title: School of Graduate Studies – Travel Award
Amount: 500 USD
Institution: University of Minnesota Duluth. August 2019

Type: Academic Scholarship
Title: Rocinante Scholarship - International student
Amount: 2,500 USD
Institution: University of Minnesota Duluth. October 2019

Type: Academic Scholarship
Title: Rocinante Scholarship - International student
Amount: 2,500 USD
Institution: University of Minnesota Duluth. October 2018

Type: Academic Scholarship
Title: International Student **Full Tuition** Scholarship
Amount: 80,000 USD
Institution: University of Minnesota Duluth. August 2014 -2018

Type: Academic Excellence Award
Title: T.O. Odlaug Outstanding Senior Biology Student Award
Institution: University of Minnesota Duluth. April 2018

Type: Academic Scholarship
Title: Stephen Welsh Scholarship - College of Liberal Arts
Amount: 3,000 USD
Institution: University of Minnesota Duluth. April 2019

Type: Academic Scholarship
Title: Western Union Scholarship
Amount: 5,000 USD
Institution: The Western Union Bank Foundation. May 2017